Numerical study and decomposition of magnetic nanowire networks

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Landau-Lifschitz equation

$$\frac{\partial m}{\partial t} = -m \wedge h_e(m) - \alpha\, m \wedge (m \wedge h_e(m)) \text{ sur } \mathcal{R}$$
$$h_e(m) = A\Delta m + h_d(m) + H_a$$

- m préserve la contrainte : $|m| = 1$
- Precession term :$-m \wedge h_e(m)$ fait tourner m autour de $h_e(m)$:
- Damping terme : $-m \wedge (m \wedge h_e(m))$ aligne m avec $h_e(m)$:

for more details, refer to [1]

Non-overlapping Schwarz Method

1. Divide the spatial domain $\mathcal{R}$ (the wire network) into non-overlapping sub-networks $\mathcal{R}_1, \dots, \mathcal{R}_p$, each handled by a different MPI process;
2. Each domain $\mathcal{R}_i$ is responsible for the evolution of m_n^i on its set of wires;
3. Each MPI process sends interface values to its neighboring processes;
4. Each MPI process reconstructs the magnetization at the connected junctions.

Example of Connected Subnetworks of Magnetic Nanowires

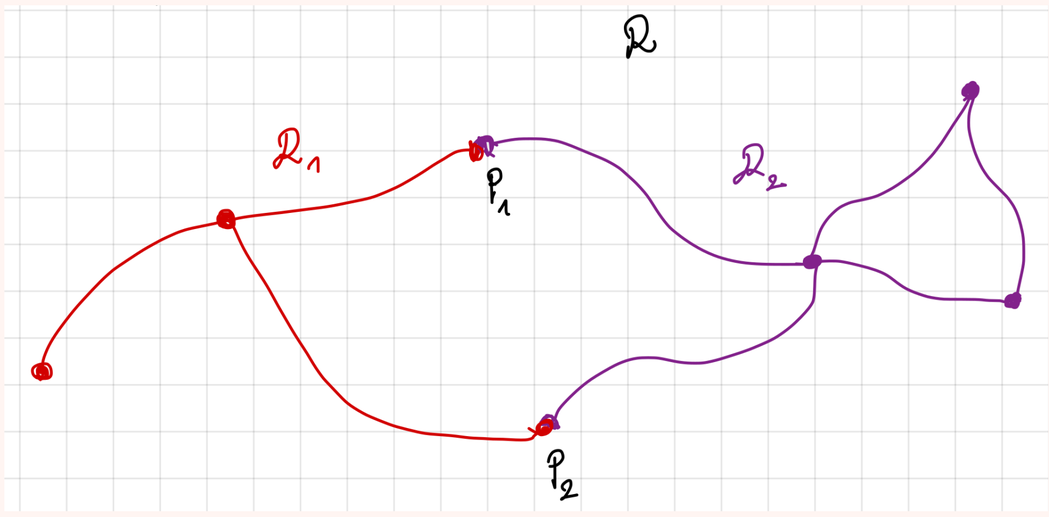


Figure 1. Decomposition of the network $\mathcal{R}$ into connected subnetworks.

Decomposition of a Network of Magnetic Nanowires into Connected Subnetworks

$$\mathcal{R} = \bigcup_{i=1}^{p} \mathcal{R}_i$$

Let $\mathcal{R}_i^0$ be a finite set of objects with Hausdorff dimension zero (points).

Let $\mathcal{R}_i^1$ be a finite set of objects with Hausdorff dimension one (curves).

We define:

$$\mathcal{R}_i = \mathcal{R}_i^0 \cup \mathcal{R}_i^1 \quad \text{where} \quad \mathcal{R}_i \subset \mathbb{R}^3$$

1. For each $\ell \in \mathcal{R}_i^1$, there exists a mapping $\varphi_\ell^i \in \mathcal{C}^\infty([0, L_\ell]; \mathbb{R}^3)$ such that for all $i \in \{1, \dots, p\}$:
$$\ell = \varphi_\ell^i([0, L_\ell])$$
2. For each $\ell \in \mathcal{R}_i^1$, there exists a pair of points $(p_{0,\ell}, p_{L_\ell,\ell}) \in (\mathcal{R}_i^0)^2$ such that:
$$\varphi_\ell^i(0) = p_{0,\ell} \quad \text{and} \quad \varphi_\ell^i(L_\ell) = p_{L_\ell,\ell}$$
3. For all $\ell \in \mathcal{R}_i^1$, we have $\ell_k \cap \ell_{k-1} \in \mathcal{R}_i^0$ for every $k \in \{1, \dots, N_i - 1\}$, where N_i is the number of wires in the subnetwork $\mathcal{R}_i^1$.
4. For each $\ell \in \mathcal{R}_i^1$, we define the set of boundary values at its endpoints as:
$$\nu_0^i(p) = \{\varphi_\ell^i(0), \varphi_\ell^i(L_\ell)\}.$$
5. We define $\nu_1^i(p)$ as the set of nanowires in $\mathcal{R}_i$ connected to the point p:
$$\forall p \in \mathcal{R}_i^0, \quad \nu_1^i(p) = \left\{\ell \in \mathcal{R}_i^1 \mid p = \varphi_\ell^i(0) \text{ or } p = \varphi_\ell^i(L_\ell)\right\}$$
6. The intersection $\mathcal{R}_i \cap \mathcal{R}_j$ is defined as the set of connection points between the subnetworks $\mathcal{R}_i$ and $\mathcal{R}_j$:
$$\mathcal{R}_i \cap \mathcal{R}_j = \left\{p \in \mathcal{R}_i^0 \cap \mathcal{R}_j^0\right\}$$
which means that the networks $\mathcal{R}_i$ and $\mathcal{R}_j$ intersect at the points p.
7. We define the set of free boundary points of the subnetwork $\mathcal{R}_i$ as:
$$\partial\mathcal{R}_i = B_i \setminus (\mathcal{R}_i \cap \mathcal{R}_j), \quad \text{where } B_i = \{p \in \mathcal{R}_i^0 \mid \#\nu_1^i(p) = 1\}.$$
8. For any function $m^p \in \mathcal{C}^1([0,T], W_{N^p})$, we define R_{N^p} as the lifting function of the network $\mathcal{R}_p$, based on Lagrange interpolation:
$$R_{N^p}(m^p) = \sum_{i=1}^{N_F^p} R_{N_p^i}(m^{p,i}) = m^p$$
with $W_{N^p} = R_{N^p}((\mathbb{R}^3)^{\sum_{i=1}^{N^p} n_i^p})$, where n_i^p is the number of discretization points in wire i of the network $\mathcal{R}_p$.
The global lifting function over the full network is given by:
$$R_N(m) = \sum_{i=1}^{p} R_{N^p}(m^p).$$

Numerical Study of Subnetworks of Magnetic Nanowires

Let $\mathcal{R}_i$ be a given subnetwork. We consider the Landau-Lifshitz equation defined for every $m^{n,i,k_i} \in H^1([0,\eta] \times \mathcal{R}_i, S^2)$, where $k_i \in \{1, \dots, N_i\}$, as follows:

$$\begin{cases}
\dfrac{\partial m^{n,i,k_i}}{\partial t} = -m^{n,i,k_i} \times h_e^{i,k_i}(m^{n,i,k_i}) - \alpha_{k_i}^i m^{n,i,k_i} \times \left(m^{n,i,k_i} \times h_e^{i,k_i}(m^{n,i,k_i})\right) & \text{on } \mathcal{R}_i \\
h_e^{i,k_i}(m^{n,i,k_i}) = A\Delta m^{n,i,k_i} + h_d^{i,k_i}(m^{n,i,k_i}) + H_a^i & \\
m^{i,k_i}(\cdot, 0) = m_0^{k_i} & \text{on } \mathcal{R}_i \\
\frac{\partial m^{n,i,k_i}}{\partial \eta^{k_i}} = 0 & \text{on } \partial\mathcal{R}_i \\
B_{i,p}(m^{n,i}) + B_{j,p}(m^{n-1,j}) = 0 & \text{on } \mathcal{R}_i \cap \mathcal{R}_j \\
B_{i,p}(m^{n,i}) = \displaystyle\sum_{k_i \in \nu_1^i(p)} \alpha_{k_i}^i \frac{\partial m^{n,i,k_i}(p)}{\partial \eta^{k_i}} & \\
m^{n,i,k_i}(p) = m^{n-1,j,k_j}(p) & \text{on } \mathcal{R}_i \cap \mathcal{R}_j
\end{cases}$$

By choosing an explicit Euler time-stepping scheme with finite differences, the solution on the subnetwork $\mathcal{R}_i$ is computed as:

$$m_{n+1}^i = m_n^i - dt\left(m_n^i \times h_e^i(m_n^i) + m_n^i \times \left(m_n^i \times h_e^i(m_n^i)\right)\right)$$

where m_{n+1}^i is the global numerical solution at time t_{n+1}, and

$$h_e^i(m_n^i) = \Delta m_n^i + \frac{1}{2}\left((m_n^i, u^i)u^i\right) + h^i(t)e_1^i,$$

with u^i the unit tangent vector of the wire and e_1^i the direction of the applied field.

To ensure the stability of the scheme, we choose an optimal time step satisfying the CFL condition:

$$dt = \frac{h_i^2}{4A}.$$

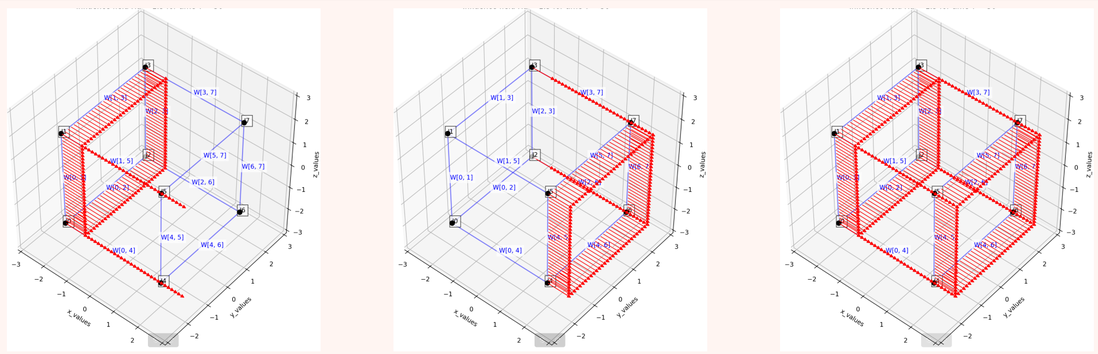
The global solution over the network $\mathcal{R}$, computed by each process at time t_{n+1}, is given by:

$$m_{n+1} = (m_{n+1}^i)_{1 \le i \le p}.$$

In other words, the finite difference scheme over the global network $\mathcal{R}$ can be rewritten as:

$$m_{n+1} = m_n - k\left(m_n \times h_e(m_n) + m_n \times (m_n \times h_e(m_n))\right).$$

Load Equilibrium among MPI Processes



(a) proc de rang 0 (b) proc de rank 1 (c) proc Maître de rang 0

Figure 2. m evolution for $H_a = 1.5$ at $t = 30$.

Comments

At each iteration, we consider $i \in \{1, \dots, n\}$, and j such that $\mathcal{R}_i$ is a neighbor of $\mathcal{R}_j$ in the domain decomposition, i.e., $\mathcal{R}_i \cap \mathcal{R}_j \neq \emptyset$.

At the junction points, the following transmission condition holds, which is key to the decomposition strategy:

$$B_{i,p}(m^n) + B_{j,p}(m_j^{n-1}) = 0.$$

To ensure convergence of the solution at the interfaces, we verify that after n time steps, there exists $\varepsilon > 0$ such that:

$$|e_n| < \varepsilon \quad \text{or} \quad e_n = \left\|B_{i,p}(m^{n,i}) + B_{j,p}(m^{n-1,j})\right\|_{H^1(\mathcal{R}_i, \mathcal{S}^2)}.$$

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